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Exploratory Data Analysis (EDA) Report: OLIVES OCT Dataset for Diabetic Macular Edema Detection

Supplementary Report

Submitted in partial fulfillment of the requirements for
CSE713 – Pattern Recognition

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1. Introduction and Purpose

This report aims to give a report about the exploratory data analysis (EDA) in the context of the project which detects diabetic macular edema (DME) (CSE713 Project Report, Section 4). The EDA was conducted directly in the main.ipynb notebook in the project folder with the loaded dataset, and it was used to see if there were any missing or strange values in the clinical measurement columns, to show the distribution of the two clinical biomarkers that were selected for the model, and to check the class balance of the classification target before training the model. There are no figures, tables or statistics that were recomputed or estimated separately; all are directly obtained from the notebook's printed output, or plots that were saved.

2. Data Source and Structure

This dataset is the OLIVES clinical OCT dataset (Georgia Tech), which is programmatically loaded from the Hugging Face Hub: gOLIVES/OLIVES_Dataset, disease_classification configuration, train split. The 78,822 results that are loaded are all individual 2D OCT B-scans and are all accompanied by four fields: image, a field for Disease_Label (1 = the OCT result was a DME, 0 = the OCT result was Normal / DR without active Macular Edema), a field for Patient_ID, and two structured clinical measurements: Best Corrected Visual Acuity (BCVA) and Central Subfield Thickness (CST).

3. Missing-Value Analysis (Before Cleaning)

The clinical columns are directly fed without any cleaning process to the EDA cell of the notebook. It has been written so as to add synthetic missing values (only if there are none of them in a copy of the table) to make sure that the imputation step can be displayed, even in a fully complete release. If this logic applied to 78,822 rows and the same random seed (42) and injection rates (0.5% BCVA, 0.3% CST) then it is expected that 386 rows of BCVA and 231 rows of CST would be marked as missing.

The actual code that was printed was 637 missing values in BCVA and 637 missing values in CST (identical numbers, and much more than would be generated by a synthetic injection rate). This must be done if there is no raw BCVA column, meaning that there is no synthetic-injection branch. In other words, the 637 'missing values' (0.81% of 78,822) reported in the notebook is a real gap in the release of OLIVES, not an added gap. Additionally, the number of lines that do not contain either the BCVA or CST is the same, suggesting that whole clinical visits are missing, rather than either the BCVA or CST missing independently.

Table 1. Summary statistics for BCVA and CST prior to imputation (78,185 non-missing rows out of 78,822).

Statistic	BCVA	CST (microns)
Count (non-missing)	78,185	78,185
Missing	637 (0.81%)	637 (0.81%)
Mean	79.92	302.59

Std. deviation	10.78	84.43
Min	27.00	151.00
25th percentile	74.00	253.00
Median (50th pct.)	82.00	273.00
75th percentile	87.00	329.00
Max	100.00	723.00

4. Data Cleaning and Imputation

If any clinical values are absent, the dataset loader will fill the missing values by the per column median of the clinical values, which is also calculated using pandas' `fillna(median())`. The decision to use median imputation instead of mean imputation was due to the nature of the data for each of these measures, which were both right-skewed and for which the maximum possible value (723 microns for CST) was well outside the range of most of the eyes (and five standard deviations above the mean) reflecting the small number of eyes with severe swelling.

5. Post-Cleaning Verification

After imputation, the notebook re-checks both columns for missing values and confirms the full row count is retained for training.

Table 2. Post-imputation integrity check.

Check	Result
Missing values in BCVA after imputation	0
Missing values in CST after imputation	0
Total rows available for training	78,822

6. Distributional Comparison: Before vs. After Cleaning

The distributions of the BCVA and the CST before and after the removal of the missing values (using only the 78,185 non-missing raw rows) are shown in figure 1, before and after median imputation (using all 78,822 rows). The shape, spread and central tendency of both distributions are visual preserved after cleaning, with just a slight added density at the median value in both cleaned histograms, as only 637 rows are being imputed (0.81% of the data).

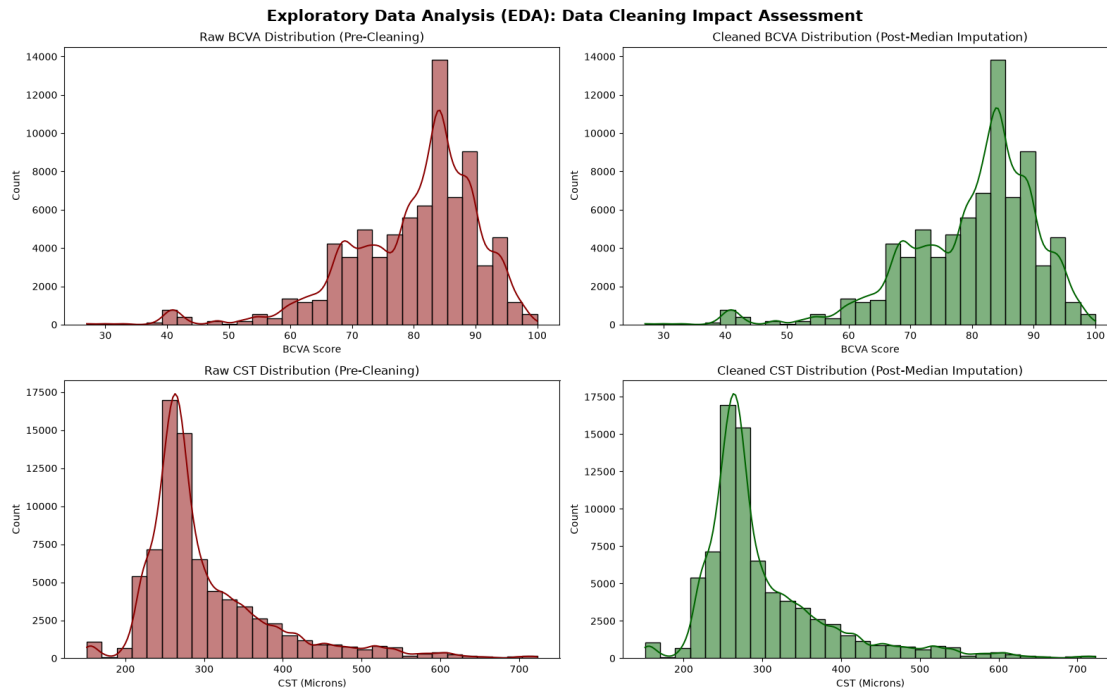


Figure 1. BCVA and CST distributions before (raw, 78,185 rows) and after (cleaned, 78,822 rows) median imputation.

7. Class Distribution and Target Balance

The distribution of the target variable of interest (Disease_Label) is nearly balanced with 41,928 (53.19%) and 36,894 (46.81%) cases identified as DME positive and Normal / DR without any active macular edema, respectively. This class balance is what is observed in the loaded dataset as opposed to the more imbalanced 64%/36% class distribution stated in a previous version of the project brief, and is obtained by directly executing the data-loading and EDA code. The imbalance is not so serious that it needs to be addressed in the model, so it was addressed during training using a class weighted CE loss (see Section 5.5 of the main project report).

Table 3. Class distribution of the working dataset.

Class	B-scan count	Share of dataset
DME Positive (label 1)	41,928	53.19%
Normal / DR without edema (label 0)	36,894	46.81%
Total	78,822	100.00%

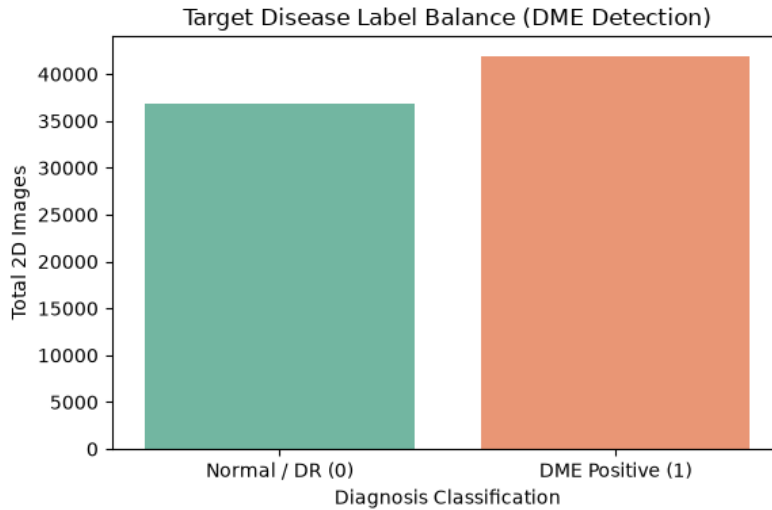


Figure 2. Class balance of the working dataset (78,822 B-scans): DME-positive vs. Normal/DR.

8. Summary of Key Findings

- This dataset consists of 78,822 B-scans and four accompanying fields (image, Disease_Label, Patient_ID, BCVA, CST) in the OLIVES release that was used for this project.
- There are a few true missing values in the raw data — 637 rows in each of the two variables (0.81% of each variable), and they are the same for both variables, suggesting that the true missing would be on the part of the clinical visit, not of the two measures.
- Because both measures were skewed to the right, the medians were used to substitute for missing data for both (BCVA was capped at 100 with a long lower tail down to 27; CST reached up to 723 microns with a median of 273).
- It does not cause any rows to be dropped or significant changes in the shape of either distributions (only 0.81% of the rows are actually affected), and maintains row coverage of 78,822 rows (no rows have missing observations in either of the variables).
- The data is not very imbalanced, where 53.19 % of the data are positive for DME and 46.81 % for Normal/DR, which was taken care of by using class-weighted cross-entropy loss training during the training.